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Class:	Data Warehouse Mini-Project	
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This mini project assignment consists of 1 task. You should focus on providing complete solution, however if you are unable to solve a particular step, try to give at least a partial solution or provide justification for the reason for the lack of a solution.

MINI-PROJECT 1 - TOPIC PROPOSAL

The process of creating a data warehouse should be preceded by an understanding of the "business needs" and reality (problem field) represented by the available data resources. The implementation of the following task is to make you aware of the problems occurring in a specific (selected) section of reality and then enable the identification (determination) of the needs, purpose, and capabilities of business analysis to support decision-making processes (making the right business decisions).

TASK - DETAILS

Please prepare the scope of the project in accordance with the specifications below, please remember that you should focus on one of the selected project datasets.

TASK - SOLUTIONS:

1.1 PROJECT TITLE

Bitcoin Blockchain Transaction Analytics - A Data Mart for On-Chain and Market Intelligence

1.2 GENERAL DESCRIPTION OF THE DOMAIN

Please introduce the selected domain. Focus on introduction of major notions, subjects and their relations.

Bitcoin is a decentralised digital currency operating on a public, immutable distributed ledger called the blockchain. The blockchain is a chronologically ordered sequence of blocks, where each block contains a set of cryptographically verified and confirmed transactions.

CORE DOMAIN CONCEPTS:

BLOCKCHAIN: The append-only ledger recording all transactions since Bitcoin's genesis block (January 3, 2009, block height 0). New blocks are appended approximately every 10 minutes by miners who solve a cryptographic Proof-of-Work puzzle. The blockchain is immutable - once a block is confirmed and buried under subsequent blocks, reverting it becomes computationally infeasible.

BLOCK: The fundamental unit of the blockchain. Each block contains:

- **Block header:** height (sequential number), hash (unique identifier), timestamp (when mined), difficulty (proof-of-work target), nonce (puzzle solution number), version (protocol version flags)
- **Block body:** typically 1,000-3,000+ transactions, including one mandatory coinbase transaction and many regular transactions
- **Metadata:** size (bytes), weight (SegWit-adjusted size metric, max 4,000,000 weight units), total fees (sum of all TX fees in the block), miner pool information

TRANSACTION (TX): An atomic event representing transfer of value. Each transaction:

- Consumes zero or more input UTXOs (Unspent Transaction Outputs) from prior transactions
- Creates one or more output UTXOs, locking value to recipient addresses via cryptographic scripts
- Includes a fee (satoshis, Bitcoin's smallest unit) paid to the miner incentivising inclusion in the next block
- Is immutable once confirmed in a block

UTXO (Unspent Transaction Output): Bitcoin's fundamental ownership tracking mechanism. Unlike traditional banking (account balances), Bitcoin uses the UTXO model: a transaction output is a tuple of (value, scriptpubkey, previous_txid, previous_output_index). An output remains "unspent" until a future transaction consumes it as an input. This UTXO set is deterministically derivable from all confirmed transactions since genesis.

SCRIPT TYPES (Locking Mechanisms):

- **P2PKH (Pay-to-Public-Key-Hash):** Legacy address format; spender must provide signature + public key
- **P2SH (Pay-to-Script-Hash):** Spender must provide a script matching the hash; enables complex conditions
- **P2WPKH (Pay-to-Witness-Public-Key-Hash):** SegWit native; witness (signature) data stored separately, reducing transaction size
- **P2WSH (Pay-to-Witness-Script-Hash):** SegWit multisig variant
- **P2TR (Pay-to-Taproot):** Latest upgrade (2021); uses elliptic curve optimisations for privacy and size reduction

MINING AND HALVINGS: Miners compete to solve the Proof-of-Work puzzle. The protocol adjusts mining difficulty every 2,016 blocks (~2 weeks) to maintain the 10-minute target block interval. Every 210,000 blocks (~4 years), the block subsidy (new Bitcoin created and awarded to the winning miner) is halved. This creates distinct economic eras:

- Era 1 (2009-2012): 50 BTC/block subsidy
- Era 2 (2012-2016): 25 BTC/block
- Era 3 (2016-2020): 12.5 BTC/block
- Era 4 (2020-2024): 6.25 BTC/block
- Era 5 (2024+): 3.125 BTC/block

As subsidies decline, miners increasingly depend on transaction fees for revenue, making fee dynamics economically critical.

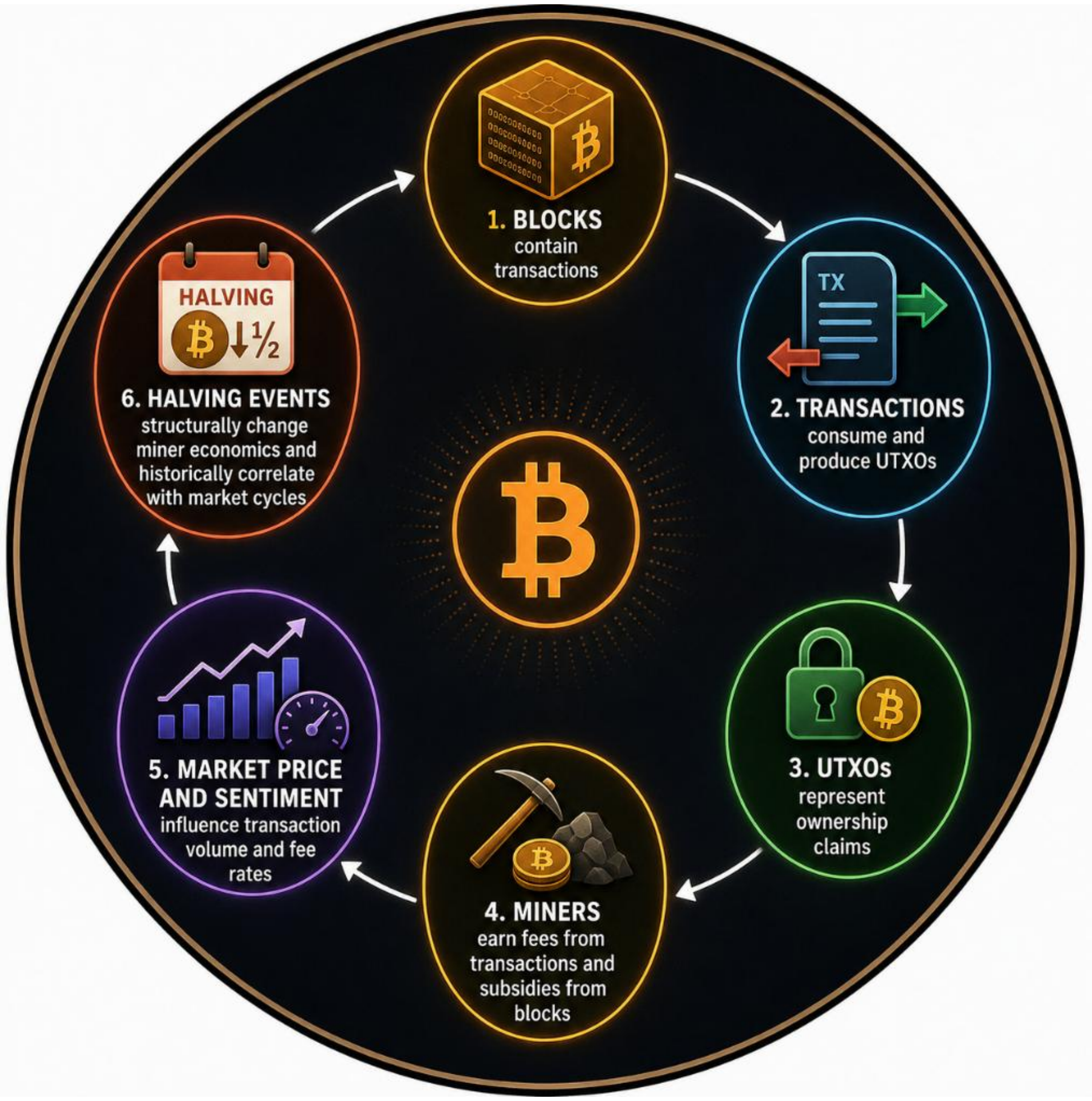
MEMPOOL: Unconfirmed transactions waiting for inclusion in a block. The mempool size, composition, and average fee rate are dynamic signals of network congestion and settlement urgency.

MARKET DATA: Bitcoin trades globally on 24/7 cryptocurrency exchanges. Its price fluctuates based on supply/demand, macroeconomic factors, and network-specific sentiment. Key market metrics include:

- OHLCV (Open, High, Low, Close, Volume) price data
- Market capitalisation (circulating supply × price)
- Fear & Greed Index (composite sentiment score, 0-100)
- Bitcoin dominance (BTC's percentage share of total crypto market cap)
- NVT Ratio (Network Value to Transactions - analogous to P/E ratio for networks)

RELATIONSHIPS:

Blocks contain transactions -> transactions consume and produce UTXOs -> UTXOs represent ownership claims -> miners earn fees from transactions and subsidies from blocks -> market price and sentiment influence transaction volume and fee rates -> halving events structurally change miner economics and historically correlate with market cycles.



1.3 DESCRIPTION OF THE POSSIBLE ANALYSIS AREAS

Please state and comment the selected fragment of the domain intended for the data warehouse. Focus on identification and introduction of the underlying possible business processes (origins of the events/states). Briefly elaborate on their granularities.

Bitcoin blockchain analytics can be broken into several distinct business processes, each with different granularity and decision contexts:

AREA 1 - TRANSACTION SETTLEMENT PROCESS

The core operational process: confirmed on-chain transactions represent discrete, measurable peer-to-peer value transfer events. Grain: one confirmed transaction per block (~10 minutes).

Analytical needs: fee dynamics, transaction composition, protocol adoption (SegWit uptake), UTXO flow, address clustering, temporal patterns.

Decision context: wallet developers, blockchain analysts, miners, traders estimating settlement costs and speed.

SELECTED FOCUS FOR THIS DATA MART

AREA 2 - BLOCK PRODUCTION PROCESS

Mining perspective: each mined block is a production event creating new Bitcoin supply and awarding fees to miners. Grain: one block per ~10 minutes.

Analytical needs: miner revenue, block fullness, difficulty adjustments, pool concentration, inter-block timing, subsidy burndown.

Decision context: miners (revenue forecasting), protocol researchers, macro Bitcoin analysts.

AREA 3 - MEMPOOL CONGESTION AND FEE MARKET PROCESS

Real-time operational process: unconfirmed transaction queue. Snapshots of pending transaction count, fee rate distribution represent transient state. Grain: one mempool snapshot per interval (per block, per hour, per minute for real-time systems).

Analytical needs: fee estimation, congestion prediction, urgency tiers, confirmation time distributions.

Decision context: exchange operators, traders needing immediate fee guidance, high-frequency settlement systems.

AREA 4 - MARKET PRICE AND SENTIMENT DYNAMICS

Financial time series: daily OHLCV price, sentiment indices. Grain: one market observation per calendar day.

Analytical needs: price-on-chain correlations, sentiment regimes, volatility clustering, cycle identification.

Decision context: traders, macro analysts, portfolio managers correlating on-chain activity with price action.

AREA 5 - UTXO LIFECYCLE AND COIN AGE

Micro-level tracking: creation and consumption of UTXOs, coin age, dormancy metrics. Grain: one UTXO state transition per transaction output.

Analytical needs: whale movement, consolidation patterns, loss detection (long-dormant coins), supply metrics.

Decision context: advanced on-chain forensics, supply-side analysis researchers.

SELECTED SCOPE: Primary focus on Area 1 (Transaction Settlement), enriched with Area 4 (Market Data) context. Area 2 attributes are captured via DIM_BLOCK (halving_era, block regime tiers) and degenerate dimension block_height. Areas 3 and 5 are deferred to future phases.

1.4 PROJECT GOAL

Please state and comment the identified main business process being the focus of the project. Identify main decision problem(s) associated with it. Provide results in section 1.4.1.

For the identified main business process, please identify basic users (types) - provide brief description of each type - and state expectations of each type. Do not select too many types - as it might be beneficial to focus on more details for 2-3 user types. Further present detailed needs by specify 10-15 OLAP user query types (do not write any SQL queries, these should be formulated in natural language - strictly in business terms). Note - these should not be very specific queries (using specific attribute values), but rather general query/need types. For each query/need please indicate how a user can utilise resultant information. Match each query to the identified user types. Provide results in section 1.4.2.

Sample inquiries:

- *How have total sales of our products evolved over the past five years, by region and sales channel?*
    - *Usage: The analysis allows you to identify the regions and channels that are most profitable or need support. Based on this, decisions can be made to increase marketing investments in high-potential regions or restructure operations in less profitable areas.*
- *Which products generated the most profit in the current quarter, and which in the same period last year?*

    - *Usage: This allows you to quickly capture changes in customer preferences and sales effectiveness. Decision makers can decide to expand the range of best-selling products or withdraw those that are loss-making.*

- *Which customers make the most purchases and what is their demographic structure (e.g. age, gender, location)?*
    - *Usage: Allows you to create personalized offers and loyalty campaigns, increasing the chances of retaining key customers and increasing the value of their shopping cart.*

Perform basic analysis of these query types (1.4.2), try to infer some general user requirements. Focus on identifying event(s) and perspectives that the user is interested in. Further, perform basic analysis of user query types (1.4.2), try to infer some detailed user requirements. List all possible measures and all possible individual dimensions. Provide results in section 1.4.3

1.4.1 THE FOCUS OF THE PROJECT

MAIN BUSINESS PROCESS: Bitcoin Transaction Settlement

Each confirmed Bitcoin transaction represents an atomic measurable event within the larger blockchain settlement system. The process captures the full lifecycle from transaction creation through confirmation in a mined block, including:

- Transaction initiation (fee selection, input/output structure)
- Mempool queueing (unconfirmed state)
- Block inclusion (confirmation)
- Finality (multiple blocks deep in chain)

CENTRAL DECISION PROBLEMS:

1. FEE MARKET DYNAMICS AND SUSTAINABILITY

Question: How do transaction fee levels and distribution evolve over time, and are transaction fees becoming a sustainable long-term revenue source for miners as subsidies halve?

Business context: As Bitcoin's block subsidy decreases, miners increasingly depend on transaction fees. Understanding fee market maturation is critical for assessing long-term network security and miner profitability.

2. PROTOCOL ADOPTION AND EFFICIENCY

Question: How quickly is the Bitcoin network adopting newer protocol upgrades (SegWit, Taproot) and script types, and what are the measurable efficiency gains?

Business context: Protocol upgrades improve scalability, privacy, and developer experience. Tracking adoption rates informs wallet developers and exchange operators about backward compatibility windows and optimal script type recommendations.

3. ON-CHAIN LEADING INDICATORS AND PRICE CORRELATION

Question: Does on-chain transaction activity (volume, fee rates) exhibit predictive power for subsequent Bitcoin price movements, or correlation patterns that can inform trading and risk management?

Business context: Traders and portfolio managers seek to identify whether on-chain metrics (settlement volumes, active addresses) lead, lag, or coincide with price trends - potentially enabling alpha generation or risk signal detection.

4. MINER ECONOMICS ACROSS HALVING EPOCHS

Question: How have miner incentives (subsidy + fees), block fullness, and transaction composition changed across halving eras, and what do these reveal about network health and miner behaviour?

Business context: Macro analysts and protocol researchers assess whether Bitcoin's economic model remains sound and whether the subsidy schedule produces expected incentive structures.

5. MARKET SENTIMENT AND USER BEHAVIOUR

Question: Do transaction volumes, fee preferences, and user urgency vary predictably across different market sentiment regimes (fear vs. greed), and can sentiment-based fee models improve transaction pricing?

Business context: Exchange operators and wallet providers can optimise fee recommendation engines by understanding whether sentiment regimes influence user behaviour, enabling dynamic pricing that balances revenue and user experience.

1.4.2 EXPECTATIONS AND DETAILED NEEDS FOR DECISION SUPPORT

USER PROFILE 1 - BLOCKCHAIN ANALYST / RESEARCHER [PRIMARY]

Description: Academics, cryptocurrency research firms, protocol researchers, and data analysts studying Bitcoin's on-chain economics, long-term trends, and network health. Typically works with aggregated daily/weekly/monthly time series and cross-dimensional comparisons. Focused on historical analysis, structural shifts (protocol upgrades), and validation of economic theories.

Characteristics:

- Time horizon: weeks to years; primarily historical analysis
- Data granularity: daily and above (weekly, monthly, quarterly, annual aggregations preferred)
- Preferred visualisations: time series line charts, stacked bars for composition, heatmaps for matrix analysis
- Key metrics: averages, percentiles, trends, growth rates

Example analytical questions this user asks:

- How has fee market matured across halving eras?
- Is SegWit adoption accelerating or plateauing?
- What is the long-term trend in miner revenue?

USER PROFILE 2 - EXCHANGE OPERATOR / TRADER [SECONDARY]

Description: Cryptocurrency exchange operators, active traders, proprietary trading firms, and market makers. Requires recent and near-real-time signals for operational decision-making: mempool congestion, fee estimation, on-chain volume spikes, whale movement detection, settlement latency prediction.

Characteristics:

- Time horizon: minutes to days; forward-looking for next moves
- Data granularity: hourly, daily (recent data; focus on last 7-30 days)
- Preferred visualisations: real-time dashboards, anomaly alerts, candlestick charts with volume overlay
- Key metrics: current values, changes (Δ), percentiles, volatility

Example analytical questions this user asks:

- What is current mempool congestion and recommended fee?
- Are there unusual on-chain settlement volumes today?
- Has market sentiment shifted in the past 24 hours?

OLAP QUERY TYPES AND BUSINESS USAGE:

Q1 - ANALYST - Fee Rate Evolution by Halving Era and Time

Question (natural language): "How has the average transaction fee rate (satoshis per virtual byte) evolved month-over-month, broken down by Bitcoin's halving eras?"

Usage: Allows analysts to assess fee market maturation across economic epochs. Reveals whether:

- Fees are increasing (indicating rising demand and potential scalability pressure)
- Fees stabilise post-spike (indicating market equilibrium)
- Fee levels are sustainable to support miner revenue as subsidies decline

Decision outcome: Informs long-term network security assessments and miner economic sustainability analyses. Protocol researchers use this to recommend blocksize or fee market adjustments.

Q2 - ANALYST - SegWit and Legacy Script Type Adoption by Quarter

Question: "What percentage of confirmed transactions use SegWit-native inputs (P2WPKH, P2WSH, P2TR) versus legacy script types (P2PKH, P2SH), and how has this composition evolved by quarter?"

Usage: Measures the pace of protocol upgrade adoption post-2017 SegWit activation and post-2021 Taproot activation. Reveals:

- Wallet and exchange support for modern script types
- User willingness to migrate from legacy formats

- Protocol upgrade success (measured by real adoption, not just deployment)

Decision outcome: Wallet developers determine when legacy support can be deprecated. Exchanges plan infrastructure migration roadmaps. Protocol researchers evaluate soft fork adoption velocity.

Q3 - TRADER - On-Chain Volume vs. Price Movements

Question: "Do on-chain confirmed transaction volumes spike in the days preceding or following large Bitcoin price movements (>5% daily change)?"

Usage: Examines whether on-chain settlement activity is a leading, lagging, or coincident indicator of price moves. If spikes precede moves:

- Traders can use volume spikes as early warning signals
- Portfolio managers can use on-chain data to validate market moves
- Risk models can incorporate on-chain signals

Decision outcome: Traders develop algorithmic rules using on-chain data. Exchange operators improve settlement capacity planning by anticipating demand spikes.

Q4 - TRADER - Fee Rate Distribution by Market Sentiment

Question: "What is the fee rate distribution (median, percentiles, spread) during Extreme Fear vs. Extreme Greed sentiment periods?"

Usage: Reveals whether user urgency correlates with sentiment. During fear periods:

- Users may be less urgent -> willing to wait for low fees
- Or: users panic-selling -> willing to pay high fees for speed

During greed periods:

- FOMO (fear of missing out) might drive urgency and higher fees
- Or: calm conditions allow patient, low-fee transactions

Decision outcome: Exchange fee engines dynamically adjust recommendations based on sentiment. Traders anticipate fee spikes during high-emotion periods.

Q5 - ANALYST - Miner Revenue Trends by Halving Era

Question: "How does total miner revenue per block (block subsidy + transaction fees) change per halving era, and what is the ratio of fee revenue to subsidy revenue over time?"

Usage: Directly addresses the critical question: "Can miners remain secure on fees alone as subsidies approach zero?" Reveals:

- Fee trends relative to subsidy decline
- Slope at which fee revenue must grow to replace subsidies
- Risk: if fee revenue does not grow proportionally, security budget shrinks

Decision outcome: Long-term viability assessments. Investment theses about Bitcoin's economics. Protocol discussions about fee markets and blocksize capacity.

Q6 - ANALYST - Transaction Size Efficiency by Script Type

Question: "What is the average transaction size in bytes and weight units, broken down by script type (P2PKH, P2SH, P2WPKH, P2WSH, P2TR), and how has this changed over time as SegWit and Taproot adoption grew?"

Usage: Quantifies on-chain efficiency gains from protocol upgrades. SegWit introduced weight units (4× discount for witness data) reducing effective transaction size. Taproot further compressed signature size. Reveals:

- Actual savings per transaction type
- Aggregate block space savings across the network
- Validation of theoretical upgrade benefits

Decision outcome: Capacity planning analyses. Protocol upgrade impact assessment. Fee projection models accounting for efficiency gains.

Q7 - ANALYST - Market Capitalisation Tier vs. Transaction Volume

Question: "What is the relationship between Bitcoin's market capitalisation tier (Micro/Small/Mid/Large) and daily transaction count? Does on-chain activity lead or lag market valuation?"

Usage: Tests network value vs. transaction volume (NVT ratio logic). If market cap increases while transaction volume stagnates:

- Price appreciation may not be justified by fundamentals
- Or: network effect improving faster than transaction volume (legitimate)

If transaction volume increases first, then price follows:

- On-chain activity is a leading indicator

Decision outcome: Valuation model development. Bubble/overvaluation detection. Fundamental vs. sentiment-driven price assessment.

Q8 - ANALYST - Block Fullness and Fee Correlation by Era

Question: "During which halving era were blocks most consistently full (by weight units), and how did this correlate with fee rate levels?"

Usage: Examines supply-demand dynamics. Full blocks = high demand = pressure for higher fees. Reveals:

- Eras with fee pressure (scarce block space)
- Eras with slack capacity (abundant space)
- Relationship between fullness and fee sustainability

Decision outcome: Blocksize debates (e.g., should max block weight be increased?). Fee market health assessment. Capacity planning.

Q9 - TRADER - Large Transactions by Volatility Regime

Question: "How does the number and total value of large transactions (e.g., >1 BTC output value) vary across different market volatility regimes (Low/Medium/High/Extreme)?"

Usage: Large transactions often represent whale activity (institutional moves, exchanges rebalancing, major transfers). Correlation with volatility reveals:

- Do whales move when markets are calm -> signalling confidence?
- Or: whales move only in high-volatility periods -> signalling distress/rebalancing?

Decision outcome: Whale activity monitoring. Institutional positioning intelligence. Liquidity risk assessment. Anomaly detection.

Q10 - ANALYST - Coinbase Transactions and Miner Supply Rewards

Question: "What is the proportion of coinbase transactions relative to total transactions over time, and how has the coinbase output value (subsidy + fees) trended across halving eras?"

Usage: Directly tracks miner reward evolution. Coinbase transactions are the sole mechanism of new Bitcoin creation. Analysis reveals:

- Supply inflation rate (new BTC per block)
- Miner fee revenue vs. subsidy breakdown
- Scheduled supply reduction (halvings)

Decision outcome: Supply-side analyses. Inflation metrics. Miner economics. Supply shocks (halving events).

Q11 - ANALYST - NVT Signal as Valuation Indicator

Question: "How does the NVT signal (Undervalued / Fair Value / Overvalued, derived from market cap / daily TX volume) correspond to subsequent 30-day price performance?"

Usage: NVT is an on-chain valuation metric (analogous to P/E for companies). Reveals whether the metric is predictive:

- If NVT "Undervalued" -> price rises in next month? -> NVT is useful signal
- If no correlation -> NVT is not predictive -> metric should be questioned

Decision outcome: Valuation model validation. Investment decision support. Metric credibility assessment.

Q12 - TRADER - Fee Estimation by Congestion and Day-of-Week

Question: "What is the median and 75th percentile fee rate during high-congestion periods (blocks near 4M weight limit) vs. low-congestion periods, segmented by day of week?"

Usage: Builds time-aware fee models. Reveals:

- Weekday patterns (e.g., higher fees on weekdays when institutional activity peaks)
- Congestion elasticity (how quickly fees rise as fullness increases)

Decision outcome: Dynamic fee recommendation engine. Exchange transaction scheduling (batch payments for low-congestion periods). User cost optimisation.

Q13 - ANALYST - Replace-by-Fee (RBF) Adoption Trends

Question: "How has the proportion of transactions signalling Replace-by-Fee (RBF) changed over time, and is RBF signalling more prevalent in specific script types or transaction size tiers?"

Usage: RBF allows users to increase fees on stuck transactions by replacing them with higher-fee versions. Adoption reflects user sophistication and mempool management maturity. Reveals:

- User understanding of fee markets
- Effectiveness of RBF as UX feature
- Fee market friction points

Decision outcome: Wallet UX improvements. Fee market analysis. Mempool management protocol design.

Q14 - ANALYST - Transaction Complexity and UTXO Management

Question: "What is the distribution of input count and output count per transaction by halving era, and does this suggest increasing or decreasing transaction complexity?"

Usage: Complex transactions (many inputs, many outputs) suggest specific use cases: consolidation (many inputs), batch payments (many outputs), mixing (both). Trends reveal:

- UTXO consolidation cycles (possibly before halvings or price rallies)
- Exchange batch payment patterns

- Fee impact of complexity (complex TXs pay more fees but benefit from batch economics)

Decision outcome: UTXO management research. Exchange operations efficiency. Blockchain forensics.

Q15 - ANALYST - Bitcoin Dominance and Settlement Activity

Question: "How does Bitcoin dominance tier (Low <40% / Medium 40-55% / High >55% of total crypto market cap) correlate with on-chain transaction volume and settlement value?"

Usage: High dominance may reflect capital rotation into Bitcoin from altcoins, which should manifest in elevated on-chain settlement. Reveals:

- Whether dominance changes correlate with on-chain activity
- If dominance is a leading or lagging indicator

Decision outcome: Macro crypto market timing. Capital flow analysis. Sentiment regime identification.

1.4.3 SCOPE OF ANALYSIS - ASPECTS EXAMINED

For each of the OLAP queries list which Fact and Dimensions would be needed to answer it. This should justify why specific dimensions are needed in the model diagram.

For each OLAP query type, the required Fact and Dimensions are identified below. This justifies the selection of specific dimensions in the dimensional model.

Q1 - Average fee rate evolution by month and halving era

Fact table: FACT_TRANSACTION

Measures required: fee_rate_sat_vbyte (AVERAGE)

Dimensions required: DIM_DATE (year, month, month_name for temporal axis) × DIM_BLOCK (halving_era for regime comparison)

Justification: DIM_DATE provides monthly time bucketing for temporal trend analysis. DIM_BLOCK halving_era segments analysis into distinct economic eras with different subsidy levels, enabling era-to-era comparison. No DIM_TX_TYPE needed (all TX types included in aggregate), no DIM_MARKET needed (fee is structural, not sentiment-dependent).

Q2 - SegWit vs. legacy script type adoption by quarter

Fact table: FACT_TRANSACTION

Measures required: COUNT (of transactions, grouped by script type)

Dimensions required: DIM_DATE (year, quarter for quarterly bucketing) × DIM_TX_TYPE (segwit_flag, primary_script_type for protocol classification)

Justification: DIM_DATE provides quarterly time aggregation. DIM_TX_TYPE distinguishes SegWit vs. legacy (segwit_flag = 1/0) and allows breakdown by specific script type (P2WPKH, P2WSH, P2TR vs. P2PKH, P2SH). DIM_BLOCK not needed. DIM_MARKET not needed.

Q3 - On-chain volume vs. price movements

Fact table: FACT_TRANSACTION

Measures required: output_value_sat (SUM for on-chain settlement volume), btc_price_usd_close (AVERAGE daily price)

Dimensions required: DIM_DATE (day for daily grain, allowing 5% daily price threshold filtering), DIM_MARKET (price_trend to classify price movement magnitude), DIM_BLOCK (tx_count_tier as proxy for settlement demand)

Justification: DIM_DATE day grain enables identification of specific dates with >5% price moves. DIM_MARKET price_trend (derived from 7-day MA) classifies price regimes. DIM_BLOCK tx_count_tier proxies for network settlement demand. The combination reveals whether on-chain volume spikes precede, coincide, or lag price moves.

Q4 - Fee rate distribution by Fear & Greed sentiment

Fact table: FACT_TRANSACTION

Measures required: fee_rate_sat_vbyte (PERCENTILE, AVERAGE, STDDEV for distribution analysis)

Dimensions required: DIM_MARKET (fear_greed_label for sentiment classification, fear_greed_bucket for fine bucketing), DIM_TX_TYPE (size_tier to control for transaction size confounders), DIM_DATE (week or month for temporal context within sentiment regimes)

Justification: DIM_MARKET fear_greed_label directly maps to the five sentiment categories. DIM_TX_TYPE size_tier is necessary because large transactions inherently have higher fee rates; size bucketing allows fair comparison of fee distributions across sentiment. DIM_DATE provides temporal anchoring and enables time-series comparison of fee distributions.

Q5 - Miner revenue by halving era

Fact table: FACT_TRANSACTION

Measures required: fee_satoshis (SUM for total fees; requires filtering for is_coinbase), computed_subsidy (block subsidy derived from halving_era and block_height)

Dimensions required: DIM_BLOCK (halving_era for era grouping, is_epoch_boundary to optionally highlight halving transition blocks), DIM_TX_TYPE (is_coinbase to separately identify coinbase transactions containing subsidy rewards), DIM_DATE (year for year-over-year trend within eras)

Justification: DIM_BLOCK halving_era groups transactions by economic epoch. DIM_TX_TYPE is_coinbase flags the coinbase TX (one per block) where subsidy is awarded; filtering FACT_TRANSACTION for is_coinbase=1 and aggregating fee_satoshis per block gives subsidy + fees. DIM_DATE year provides temporal granularity showing revenue trends within each era.

Q6 - Transaction size efficiency by script type over time

Fact table: FACT_TRANSACTION

Measures required: tx_size_bytes (AVERAGE), tx_weight_units (AVERAGE)

Dimensions required: DIM_TX_TYPE (primary_script_type for script classification), DIM_DATE (quarter or year for adoption timeline)

Justification: DIM_TX_TYPE primary_script_type distinguishes P2PKH, P2SH, P2WPKH, P2WSH, P2TR. DIM_DATE quarter/year enables tracking size efficiency improvements post-SegWit (2017) and post-Taproot (2021). DIM_BLOCK not needed. DIM_MARKET not needed.

Q7 - Market cap tier vs. transaction count

Fact table: FACT_TRANSACTION

Measures required: COUNT (transaction count), market_cap_usd (AVERAGE of daily market cap)

Dimensions required: DIM_MARKET (market_cap_tier for market size classification), DIM_DATE (month for monthly grain enabling 30-day rolling windows and lead/lag analysis)

Justification: DIM_MARKET market_cap_tier bucketed from computed market cap value. DIM_DATE month enables comparison across market cycles and identification of lead/lag correlations. DIM_TX_TYPE and DIM_BLOCK not strictly required.

Q8 - Block fullness and fee correlation by era

Fact table: FACT_TRANSACTION

Measures required: fee_rate_sat_vbyte (AVERAGE, PERCENTILE), tx_count (implicit in DIM_BLOCK.tx_count_tier)

Dimensions required: DIM_BLOCK (size_tier and tx_count_tier for fullness classification, halving_era for era context), DIM_DATE (year, quarter for temporal trend)

Justification: DIM_BLOCK size_tier and tx_count_tier directly encode block fullness (large blocks, high TX count = full). DIM_BLOCK halving_era segments by era. DIM_DATE provides time axis. The combination reveals fee pressure during full-block periods across eras.

Q9 - Large transactions by volatility regime

Fact table: FACT_TRANSACTION

Measures required: COUNT (of large transactions), output_value_sat (SUM)

Dimensions required: DIM_MARKET (volatility_regime for volatility classification), DIM_TX_TYPE (size_tier to define "large"), DIM_DATE (day or week for granularity)

Justification: DIM_MARKET volatility_regime (Low/Medium/High/Extreme, derived from 14-day realised volatility) classifies market conditions. DIM_TX_TYPE size_tier "XL 4000+b" or output_value_sat threshold defines "large". DIM_DATE allows whale activity detection on specific dates.

Q10 - Coinbase transactions and miner reward trends

Fact table: FACT_TRANSACTION

Measures required: COUNT (coinbase TXs per block), fee_satoshis (SUM via coinbase TXs), output_value_sat (SUM of coinbase rewards = subsidy + fees)

Dimensions required: DIM_TX_TYPE (is_coinbase = 1 to filter), DIM_BLOCK (halving_era for subsidy context), DIM_DATE (year for supply trend)

Justification: DIM_TX_TYPE is_coinbase directly identifies the one coinbase TX per block. DIM_BLOCK halving_era shows how subsidy changed across eras. DIM_DATE year shows supply over time. DIM_MARKET not needed.

Q11 - NVT signal vs. price performance

Fact table: FACT_TRANSACTION

Measures required: market_cap_usd, output_value_sat (for computing NVT = market_cap / daily_TX_volume), btc_price_usd_close

Dimensions required: DIM_MARKET (nvt_signal for NVT valuation bucket, price_trend for subsequent 30-day price bucket), DIM_DATE (month for monthly NVT observation, enabling lead-lag analysis)

Justification: DIM_MARKET nvt_signal (Undervalued / Fair Value / Overvalued) is the independent variable. DIM_MARKET price_trend (30 days hence) is the dependent variable. DIM_DATE enables time-lagged correlation analysis. The query joins FACT rows from dates D with price_trend rows from dates D+30.

Q12 - Fee rates by congestion and day-of-week

Fact table: FACT_TRANSACTION

Measures required: fee_rate_sat_vbyte (MEDIAN, PERCENTILE_CONT(0.75))

Dimensions required: DIM_BLOCK (tx_count_tier for congestion classification), DIM_DATE (day_of_week for weekday pattern)

Justification: DIM_BLOCK tx_count_tier bucketed as (Low <500 / Medium 500-2000 / High >2000) serves as congestion proxy. DIM_DATE day_of_week (Monday-Sunday) reveals intra-week patterns. DIM_MARKET and DIM_TX_TYPE not required.

Q13 - RBF adoption trends by script and size

Fact table: FACT_TRANSACTION

Measures required: COUNT (transactions with is_rbf_signalling = 1)

Dimensions required: DIM_TX_TYPE (is_rbf_signalling flag, primary_script_type, size_tier to enable breakdown by script type and size class), DIM_DATE (quarter, year for adoption timeline)

Justification: DIM_TX_TYPE is_rbf_signalling directly identifies RBF-flagged TXs. Breakdown by primary_script_type and size_tier reveals whether RBF is more prevalent in specific TX types. DIM_DATE tracks adoption trajectory post-RBF standardisation (2015). DIM_BLOCK and DIM_MARKET not needed.

Q14 - Input/output count complexity by era

Fact table: FACT_TRANSACTION

Measures required: COUNT (distribution of TXs), average input/output per TX

Dimensions required: DIM_TX_TYPE (input_count_bucket and output_count_bucket for complexity classification), DIM_BLOCK (halving_era for era context), DIM_DATE (year for trend)

Justification: DIM_TX_TYPE input/output buckets classify complexity directly (1 input vs. 2-5 vs. 6-20 vs. 20+). DIM_BLOCK halving_era segments by epoch. DIM_DATE year shows whether complexity is increasing/decreasing over time. DIM_MARKET not needed.

Q15 - Bitcoin dominance vs. settlement activity

Fact table: FACT_TRANSACTION

Measures required: COUNT (TX count as settlement activity proxy), output_value_sat (SUM for settlement volume), btc_dominance_pct (AVERAGE)

Dimensions required: DIM_MARKET (btc_dominance_tier for dominance classification, cycle_phase for additional market context), DIM_DATE (month for monthly alignment)

Justification: DIM_MARKET btc_dominance_tier (Low <40% / Medium 40-55% / High >55%) classifies dominance regimes. DIM_MARKET cycle_phase (Accumulation / Early Bull / Late Bull / Distribution / Bear) provides additional market regime context. DIM_DATE month aligns on-chain activity with dominance observations. DIM_TX_TYPE and DIM_BLOCK not required.

1.5 DATA SOURCES

Prepare a brief description of the selected data source - focus on general structure and its usage, access options, update characteristics, volume, time span for facts, etc. Further perform initial assessment of the quality of available data - focus on number of records, number of valid records, number of missing records.

1.5.1 LOCATION, FORMAT, AVAILABILITY

SOURCE 1 - Bitcoin On-Chain Data

Location: Public REST APIs

- mempool.space: **https://mempool.space/api/v1/blocks**, **https://mempool.space/api/block/{hash}/txs/{page}**
- blockchain.com: https://blockchain.com/api/blockchain_api (alternative, similar endpoints)

Format:

- Live data: JSON REST API responses
- Historical archival: CSV flat files exported from above APIs or local node

Authentication: None required for public mempool.space and blockchain.com endpoints. Bitcoin Core RPC requires node setup with credentials.

Update rate: ~10 minutes per new block (blocks added to chain approximately every 10 minutes on average)

Data availability:

- mempool.space: continuous, historical data available
- blockchain.com: continuous, historical data available
- Uptime: >99.5% for both public APIs; maintained by community services with high availability

Practical access method for this project: Python requests library with automatic retries and rate-limiting; yfinance Python library for market data (no authentication needed).

SOURCE 2 - Bitcoin Market Data

Location: Multiple public APIs + Python library

- Yahoo Finance: yfinance Python package (no API key required) - https://finance.yahoo.com/quote/BTC-USD
- Fear & Greed Index: https://api.alternative.me/fng/
- Bitcoin Dominance: https://api.coinlore.net/api/global/ (live data only; historical from embedded static table)

Format:

- yfinance: Python pandas DataFrame (seamless in-process data access)
- Alternative.me: REST JSON API
- CoinLore: REST JSON API
- Embedded table: CSV/Python dict (static historical dominance data)

Authentication: None required for any of the above

Update rate:

- OHLCV price data (yfinance): daily candles (updates after market close UTC)
- Fear & Greed (Alternative.me): daily (new score published each day at fixed UTC time)
- BTC dominance (CoinLore): live endpoint provides current value; historical dates sourced from static monthly interpolation table

Data availability:

- yfinance BTC-USD: available from ~mid-2010 onwards (first reliable exchange data)
- Fear & Greed: available from 2019-02-01 onwards (no pre-2019 history)
- Dominance: live from CoinLore; monthly anchor points from 2009-2026 in static embedded table

Practical scope decision: For this project, the selected 4-year window (2021-2025) falls entirely within Fear & Greed coverage (post-2019), eliminating data availability constraints for the practical scope.

1.5.2 DATA SOURCE BASIC INFORMATION

	Source	Number of rows	Number of attributes	Size	Update rate	Grain
1	S1: Blocks (mempool)	~200K blocks	14 (block header + extras)	10-50 GB raw	~10 min (per block)	One block per ~10 minutes
2	S1: Transactions	~50M (sampled subset)	12 (tx metadata)	(raw JSON/CSV)	~10 min (per block)	One transaction per event
3	S1: Inputs/Outputs	~100M entries (filtered)	4 (value, script, address, witness)	(nested in JSON)	~10 min	One I/O per TX output/input
4	S2: OHLCV (yfinance)	~5,500 rows	6 (date, open, high, low, close, volume)	~2 MB	Daily	One market observation per calendar day
5	S2: Fear & Greed	~2,000 rows	2 (score, label)	~100 KB	Daily	One sentiment observation per day
6	S2: Dominance (static + live)	~12 rows/year historical; 1 live	2 (date, pct)	~10 KB	Monthly (static); live (CoinLore)	One dominance snapshot per calendar day (interpolated)

1.5.3 DATA SOURCE DETAILS

Study the selected dataset and present your findings in a form of a domain data dictionary (with information about table/sheet/location, attribute name, attribute type (high level type representation, like Numerical, Money, Text, Date, etc.), description (short description of the meaning of the attribute)) and quality assessment sheet (with information about table/sheet/location, attribute name, attribute type (lower-level type representation, like varchar(20), decimal(5,2), etc.), type of data (nominal, ordinal, interval, ratio, continuous), number of unique values, null ratio, quality assessment description (short description of the results of the attribute quality assessment - focusing on a column consistency assessment)).

DOMAIN DATA DICTIONARY

DOMAIN DATA DICTIONARY - Source 1 (On-Chain Data)

	Location	Attribute name	Attribute type	Description
1	height	Numerical	Sequential block number (0 = genesis); no gaps	blocks endpoint
2	hash	Text	64-character hexadecimal block identifier (unique)	blocks endpoint
3	timestamp	Date/Time	Unix epoch timestamp when block was mined	blocks endpoint
4	size_bytes	Numerical	Total block size in bytes (raw transaction data)	blocks endpoint
5	weight_units	Numerical	SegWit-adjusted weight (max 4,000,000 units); NULL for pre-SegWit blocks	blocks endpoint
6	difficulty	Numerical	Mining difficulty target (increases over time, adjusted every 2,016 blocks)	blocks endpoint
7	nonce	Numerical	32-bit proof-of-work solution number	blocks endpoint
8	tx_count	Numerical	Number of transactions in the block (includes coinbase)	blocks endpoint
9	total_fees_sat	Numerical	Sum of all transaction fees in block (satoshis)	blocks endpoint
10	mediantime	Date/Time	Median timestamp of prior 11 blocks (used for time-lock calculations)	blocks endpoint
11	pool_name	Text	Mining pool operator name (e.g., "Foundry USA", "AntPool")	blocks endpoint
12	txid	Text	64-character hexadecimal transaction identifier (unique per transaction)	tx endpoint
13	block_height	Numerical	Height of block containing this transaction	tx endpoint
14	input_count	Numerical	Number of inputs (UTXOs consumed)	tx endpoint
15	output_count	Numerical	Number of outputs (UTXOs created)	tx endpoint
16	fee_satoshis	Numerical	Fee paid to miner (satoshis); 0 for coinbase TXs	tx endpoint
17	fee_rate_sat_vbyte	Numerical	Fee density: satoshis per virtual byte (computed from fee / vsize)	tx endpoint
18	is_coinbase	Boolean	TRUE if first transaction in block (creates new Bitcoin supply)	tx endpoint
19	segwit_flag	Boolean	TRUE if transaction uses SegWit encoding (witness data separated)	tx endpoint
20	tx_size_bytes	Numerical	Transaction size in bytes (legacy + witness)	tx endpoint
21	tx_weight_units	Numerical	Transaction weight units (legacy × 3 + witness × 1); NULL pre-SegWit	tx endpoint

22	script_type	Text	Dominant output script type (P2PKH, P2SH, P2WPKH, P2WSH, P2TR, OTHER)	tx endpoint
23	is_rbf_signalling	Boolean	TRUE if transaction signals Replace-by-Fee (sequence < 0xfffffffe in any input)	tx endpoint
24	value_sat	Numerical	UTXO value in satoshis	tx inputs/outputs
25	script_type	Text	Output script type locking mechanism	tx inputs/outputs
26	address	Text	Bitcoin address (if applicable/decodable; may be NULL for non-standard scripts)	tx inputs/outputs
27	has_witness	Boolean	TRUE if input has witness (SegWit indicator)	tx inputs/outputs

DOMAIN DATA DICTIONARY - Source 2 (Market Data)

	Location	Attribute name	Attribute type	Description
1	yfinance BTC-USD	date	Date	Calendar date of price observation
2	yfinance BTC-USD	price_open	Money	Opening BTC price in USD
3	yfinance BTC-USD	price_high	Money	Daily high BTC price in USD
4	yfinance BTC-USD	price_low	Money	Daily low BTC price in USD
5	yfinance BTC-USD	price_close	Money	Closing BTC price in USD
6	yfinance BTC-USD	volume_24h_usd	Numerical	24-hour trading volume in USD (aggregated across major exchanges)
7	Alternative.me	date	Date	Calendar date of sentiment observation
8	Alternative.me	fear_greed_score	Numerical	Composite sentiment score (0 = Extreme Fear, 100 = Extreme Greed)
9	Alternative.me	fear_greed_label	Text	Textual sentiment classification (Extreme Fear / Fear / Neutral / Greed / Extreme Greed)
10	CoinLore (live)	date	Date	Calendar date of observation
11	CoinLore (live)	btc_dominance_pct	Numerical	Bitcoin market cap as percentage of total crypto market cap
12	Static table (embedded)	date	Date	Calendar date (monthly anchors)
13	Static table (embedded)	btc_dominance_pct	Numerical	Monthly BTC dominance; daily values interpolated linearly
14	Computed in pipeline	market_cap_usd	Money	Bitcoin market cap computed deterministically from halving schedule: btc_circulating_supply(block_height) × price_close_usd

QUALITY ASSESMENT SHEET

QUALITY ASSESSMENT SHEET - Source 1 (On-Chain Data)

	Location	Attribute name	Attribute type	Type of data	Number of Unique values	Number of Null values	Quality assessment
1	blocks	height	INT	Ratio	~200,000	0%	EXCELLENT: Monotonically increasing, no gaps. Every block has unique, sequential height.
2	blocks	hash	CHAR(64)	Nominal	~200,000	0%	EXCELLENT: Cryptographically unique. Every hash is 64 hex characters. No duplicates.
3	blocks	timestamp	BIGINT	Interval	~200,000	0%	GOOD: Generally monotonically increasing. Occasional non-monotonic blocks due to miner clock skew allowed by protocol (±2 hours tolerance). Within acceptable protocol bounds. Use mediantime for strict ordering.
4	blocks	size_bytes	INT	Ratio	1,000-2,000	0%	GOOD: Bounded by protocol limits. Rare oversized blocks in early history (before protocol hard-coded limits). Valid range ~1 byte to ~4 MB.
5	blocks	weight_units	INT	Ratio	1,000-2,000	0%	MODERATE: NULL for all blocks before SegWit activation (height < 481,824, August 1, 2017). Post-SegWit blocks have weights bounded at 4,000,000. ETL must handle NULL as missing-by-design (pre-SegWit blocks had no weight metric).
6	blocks	difficulty	FLOAT	Ratio	1,000+	0%	EXCELLENT: Deterministically computed by protocol every 2,016 blocks. Monotonically increasing long-term (network grew stronger). No errors or anomalies.
7	blocks	tx_count	INT	Ratio	1-3,000+	0%	EXCELLENT: Directly countable from block content. Every block has ≥1 (coinbase). No missing values.
8	blocks	total_fees_sat	BIGINT	Ratio	10,000+	~0.5%	GOOD: Computable from TX data. Early blocks (2009-2010) have near-zero fees due to zero fee requirements. No anomalies. Rare NULL values for early blocks handled as 0.
9	tx	txid	CHAR(64)	Nominal	~50M (sample)	0%	EXCELLENT: Cryptographically unique per transaction. Every txid is exactly 64 hex chars. No duplicates within chain.
10	tx	fee_satoshis	BIGINT	Ratio	100,000+	~0.01%	EXCELLENT: Computable from TX inputs/outputs (output_sum - input_sum). Coinbase TXs have fee = 0 (not NULL; intentionally zero). Rare edge cases (<0.01%) in

							early blocks with unusual structures, set to NULL if uncomputable.
11	tx	fee_rate_sat_vbyte	DECIMAL (10,4)	Ratio	10,000+	~0.01%	GOOD: Derived field (fee / vbyte). Coinbase TXs have rate = 0 (not NULL). Pre-SegWit TXs lack witness weight, so vbyte approximated as size_bytes. Rare NULL (<0.01%) for non-standard structures.
12	tx	is_coinbase	BIT	Nominal	2 (True/False)	0%	EXCELLENT: Deterministically identified as the first TX in every block. Exactly one per block. Perfect data quality.
13	tx	segwit_flag	BIT	Nominal	2 (True/False)	~5%	MODERATE: NULL/absent for all blocks before SegWit activation (height < 481,824). Post-activation, accurately reflects presence of witness data. ETL must conditionally set pre-SegWit TXs to FALSE or NULL depending on design (recommend FALSE for all pre-SegWit; they are non-SegWit).
14	tx	tx_size_bytes	INT	Ratio	100-10,000+	0%	EXCELLENT: Directly available from API. Every TX has a size. No missing values.
15	tx	tx_weight_units	INT	Ratio	100-40,000	~5%	MODERATE: NULL for pre-SegWit blocks. Post-SegWit, accurately reflects weight. For analysis spanning both eras, ETL must impute pre-SegWit weight as size_bytes × 4 (no witness discount).
16	tx	script_type	VARCHAR (20)	Nominal	5-7 (P2PKH, P2SH, P2WPKH, P2WSH, P2TR, OTHER)	~2%	GOOD: Most TXs are decodable standard script types. Non-standard or exotic scripts mapped to "OTHER". Null ratio ~2% for edge cases (non-decodable scripts in early blocks).
17	tx	is_rbf_signalling	BIT	Nominal	2 (True/False)	~1%	GOOD: Detectable from input sequence numbers (sequence <= 0xfffffffd signals RBF). Small null rate (~1%) for non-standard inputs with unusual sequence values that don't fit standard RBF detection.
18	Inputs / outputs	value_sat	BIGINT	Ratio	100,000+	0%	EXCELLENT: Every UTXO has a defined value. No missing values.
19	Inputs / outputs	script_type	VARCHAR (20)	Nominal	5-7	~3%	GOOD: Most scripts are standard; edge cases mapped to "OTHER". Null rate ~3% for undecodable scripts.

QUALITY ASSESSMENT SHEET - Source 2 (Market Data)

	Location	Attribute name	Attribute type	Type of data	Number of Unique values	Number of Null values	Quality assessment
1	yfinance	date	DATE	Interval	~5,500	0%	EXCELLENT: Complete daily series from ~mid-2010 to present. No gaps in weekday/trading day sequence.
2	yfinance	price_open / high / low / close	DECIMAL (12,2)	Ratio	~5,500 each	0%	EXCELLENT: Complete OHLCV history. No missing values. Minor retroactive adjustments possible (rare, handled transparently by yfinance).
3	yfinance	volume_24h_usd	BIGINT	Ratio	~5,500	~0.5%	GOOD: Very early dates (2010-2011) may have zero or missing volume due to illiquid markets, exchange closures. Small null ratio acceptable.
4	Alternative.me	date	DATE	Interval	~2,000	0%	GOOD: Complete daily series **ONLY FROM 2019-02-01 ONWARDS**. NO DATA BEFORE THIS DATE. Significant coverage gap for pre-2019 analysis.
5	Alternative.me	fear_greed_score	INT	Ordinal	101 (0-100)	0%	GOOD: Complete for dates within coverage. Consistent numeric scale. One value per date.
6	Alternative.me	fear_greed_label	VARCHAR (15)	Nominal	5	0%	EXCELLENT: Consistent 5-category labelling (Extreme Fear / Fear / Neutral / Greed / Extreme Greed). Deterministically derived from score; never NULL.
7	CoinLore live	btc_dominance_pct	DECIMAL (5,2)	Ratio	1,000+	0%	MODERATE: Live API provides **ONLY TODAY'S VALUE**. All historical dates are resolved from embedded **STATIC MONTHLY TABLE** (2009-2026) with **LINEAR INTERPOLATION** between monthly anchor points. Interpolation introduces approximation error (~0.5-2% inaccuracy) at daily granularity.
8	Computed	market_cap_usd	DECIMAL (22,2)	Ratio	10,000+	0%	EXCELLENT: Computed deterministically from halving schedule (btc_circulating_supply(block_height) × price_close_usd). No external API dependency. Fully reproducible.

1.5.4 DATA SOURCE ASSESEMENT (ADDITIONAL INFORMATION. QUALITY FINDINGS, TIME SPAN FOR FACTS, ETC.)

TIME SPAN AND SCOPE DECISIONS:

On-chain data: Available from genesis block (January 3, 2009, block 0) to present.

Practical scope for this project: Last ~200,000 blocks (~4 years, approximately 2021-2025). This window contains ~50 million transactions (exact count depends on sampling strategy).

Rationale: Balances analytical richness (enough data for robust trends) with computational manageability for a class project.

Bitcoin OHLCV price data (yfinance): Available from ~mid-2010 (first reliable Yahoo Finance BTC-USD quotes) to present. Practical scope: same 4-year window (2021-2025).

Fear & Greed Index (Alternative.me): Available ONLY from 2019-02-01 onwards.

Critical finding: Pre-2019 transactions (before 2019-02-01) have no fear/greed sentiment data. The selected 4-year scope (2021-2025) falls entirely within the coverage period, so this is NOT a limiting constraint for the practical scope. However, extending analysis to earlier eras would require handling missing sentiment data via default DIM_MARKET members.

Bitcoin dominance (CoinLore): Live API provides only today's value. Historical daily values must be interpolated from a static monthly table (anchor points at first day of each month, 2009-2026). Interpolation between monthly points introduces minor approximation (~0.5-2% error at daily grain). For the practical scope (recent 4 years), this is acceptable.

Market cap (computed): Requires no external API. Computed as btc_circulating_supply(block_height) × price_close_usd using Bitcoin's deterministic halving schedule.

NVT Signal (post-load computation): NVT = market_cap_usd / daily_settlement_volume. Market cap is now computed; settlement volume is derived from FACT_TRANSACTION outputs. NVT computation is deferred until both values are available (post-fact load). See ETL staging layer strategy below.

KEY DATA QUALITY FINDINGS:

Finding 1: SegWit and Weight-Related Nulls

- Weight fields (weight_units, tx_weight_units) are NULL for all blocks before SegWit activation (block height < 481,824, August 1, 2017).
- Transactions in pre-SegWit blocks also lack witness data (segwit_flag = NULL).

ETL action: Pre-SegWit blocks classified as segwit_flag = FALSE (not NULL). tx_weight_units computed retroactively as size_bytes × 4 (witness data considered as 1 weight unit per legacy byte). No discount applies (pre-SegWit had no witness separation).

Finding 2: Coinbase Transaction Fee Handling

- Coinbase transactions (one per block) have fee_satoshis = 0 (they create new supply; no fees paid from inputs).
- These MUST NOT be excluded from analysis; they must be flagged via DIM_TX_TYPE.is_coinbase = TRUE.
- Failing to handle coinbase properly: (a) distorts average fee statistics (coinbase 0-fee drags down averages), (b) undercounts transaction volume.

ETL action: Stage fee_satoshis = 0 for coinbase. Filter is_coinbase = TRUE when computing fee averages for non-coinbase analysis.

Finding 3: Input and Output Value Aggregation

- API returns nested arrays (vin and vout), not flat fee totals.
- input_value_sat = SUM of all vin[].prevout.value per transaction.
- output_value_sat = SUM of all vout[].value per transaction.

ETL action: Explode JSON arrays into staging tables STG_TX_INPUTS and STG_TX_OUTPUTS. Aggregate back to transaction level during DW load using SUM(GROUP BY txid).

Finding 4: Primary Script Type (Majority Voting)

- Transactions can have mixed script types in inputs and outputs (e.g., spending P2PKH input to P2WPKH output).
- DIM_TX_TYPE.primary_script_type must be single-valued.

ETL action: Apply mode (most common script type) across all inputs/outputs per transaction during post-load aggregation. If tie, deterministic tie-breaker (e.g., lexicographic order).

Finding 5: Fear & Greed Coverage Gap (NOT blocking for practical scope)

- Alternative.me provides fear/greed data only from 2019-02-01 onwards.
- Pre-2019 transactions: FACT_TRANSACTION rows must LEFT JOIN to NULL on fear_greed_score.
- DIM_MARKET requires a default member for "Unknown Sentiment" (e.g., market_key = -1).

ETL action: LEFT JOIN on date. Null fear/greed rows -> map to market_key = -1 ("Sentiment unavailable"). For the practical 4-year scope (2021-2025), this does not apply; all rows fall within coverage.

Finding 6: Price and Volume Data Before ~Mid-2010

- yfinance BTC-USD history effectively starts around mid-2010 (first reliable exchange quotes).
- Blocks/transactions before ~mid-2010 will lack price/volume context.

ETL action: FACT_TRANSACTION rows pre-2010 will have NULL price and volume fields. This is acceptable; analysis can filter to post-2010 dates.

Finding 7: Bitcoin Dominance Interpolation

- CoinLore live API provides only today's value.
- All historical dates sourced from embedded static monthly table (anchor points on 1st of each month, 2009-2026).
- Daily historical values: linear interpolation between monthly anchors.

Approximation: Dominance swings 1-2% per month; interpolation error ~0.5-2% at daily grain.

ETL action: Embed monthly dominance table. Apply linear interpolation SQL function for historical dates. Use live CoinLore API only for today's value.

Finding 8: Transaction Sampling Strategy (Critical for Manageability)

- 200K blocks × 1000+ TX/block avg = ~50M+ transactions in full history.
- For a class project, loading all 50M TXs may be computationally prohibitive.

Recommended strategy: Stratified sampling

- Load 100% of coinbase transactions (one per block; mandatory for miner economics analysis) = ~200K rows
- Load ~N% of regular transactions from each block (e.g., every 5th TX, or 20% sample) = ~10M rows
- Ensures representation across all blocks, time periods, and script types

ETL action: Implement sampling in extract phase. Document sampling rate and ensure analysis accounts for sampled nature (e.g., aggregate volumes must be scaled up).

Finding 9: NVT Deferred Computation

- NVT requires both market_cap_usd (now computed) and daily settlement volume (from FACT_TRANSACTION).
- These are not available until FACT is fully loaded.

ETL action: Compute NVT as post-load step. Create a view or temporary table joining DIM_DATE to FACT aggregates ($\text{daily\_tx\_volume} = \text{SUM}(\text{output\_value\_sat})\ \text{per date}$), then compute $\mathrm{NVT} = \mathrm{market\_cap} / \mathrm{daily\_volume}$. Populate DIM_MARKET.nvt_signal from computed NVT.

Finding 10: Block Reorg Handling

- Bitcoin allows rare block reorgs (chain reorganisation) when competing chains reconverge.
- Reorged blocks are valid but no longer "canonical" (not on longest chain).

ETL action (defensive): Add is_active BIT DEFAULT 1 to all staging tables (STG_BLOCKS, STG_TRANSACTIONS, STG_TX_INPUTS, STG_TX_OUTPUTS). On reorg detection, set is_active = 0 for affected rows. Full deletions are not recommended (audit trail loss); soft deletes preferred. Load only is_active = 1 rows into DW.

IMPACT ON Q1.4.2 QUERY TYPES:

Queries Q4 and Q11 (sentiment-dependent): Limited to post-2019-02-01 for full sentiment coverage. The selected 4-year scope (2021-2025) avoids this limitation entirely.

Queries using price data (Q3, Q7, Q11, Q15): Limited to post-~2010 for complete price history. Again, 4-year scope is not affected.

Queries using dominance (Q15): Uses interpolated dominance values; minor (~1-2%) approximation error acceptable for trend analysis.

Queries using NVT (Q11): Requires post-load computation; available only after FACT and DIM_MARKET are populated. Not a hard blocker, just a sequencing requirement in ETL.

All 15 OLAP queries are answerable within the practical scope with identified data quality controls applied.

1.6-DIMENSIONAL MODEL SYNOPSIS

Prepare a brief description of the hypothesized dimensional model - focus on underlying business process that your users are interested in (remember we are building a data mart - a single line of business is enough (one process) and required dimensions. Highlight the general structure, i.e., name facts and measures, name dimensions, and list required individual dimensions (dimension's attributes).

For each, provide a single "Granularity Statement" section, e.g., "one row in the fact table represents [Event] for one [Dimension A] per [Dimension B] at the [Time Level]."

The dimensional model is a standard star schema centred on a single fact table (FACT_TRANSACTION) representing confirmed Bitcoin transactions. The schema is designed to support OLAP analysis of on-chain settlement activity across four analytical dimensions: time (DIM_DATE), block characteristics (DIM_BLOCK), transaction type (DIM_TX_TYPE), and market conditions (DIM_MARKET). The model encodes both structural (block, transaction composition) and market (price, sentiment) perspectives, enabling rich cross-dimensional analysis spanning 15+ OLAP query types.

Granularity Statement (General):

"One row in FACT_TRANSACTION represents one confirmed Bitcoin transaction included in a mined block, observed on one calendar day (DIM_DATE), within one block regime classification (DIM_BLOCK), belonging to one transaction type classification (DIM_TX_TYPE), and coinciding with one market condition state (DIM_MARKET)."

Degenerate dimensions (stored directly in FACT_TRANSACTION, not separate dimension tables):

- txid (CHAR(64)): Unique transaction identifier, enabling drill-through to individual transaction records
- block_height (INT): Exact block reference for transaction, preserving precise traceability without requiring a separate dimension row per block

1.6.1 FACTS AND MEASURES

	Fact	Measure(s)	Grain
1	FACT_TRANSACTION	fee_satoshis (BIGINT) - Transaction fee paid to miner [SUM, AVG]	One row = one confirmed Bitcoin transaction within a block, on a calendar day
		fee_rate_sat_vbyte (DECIMAL) - Fee density (satoshis per virtual byte) [AVG, PERCENTILE]	
		input_value_sat (BIGINT) - Total value of consumed UTXOs [SUM]	
		output_value_sat (BIGINT) - Total value of created UTXOs [SUM]	
		tx_size_bytes (INT) - Transaction size in bytes [AVG, MAX]	
		tx_weight_units (INT) - SegWit-adjusted weight [AVG, SUM]	
		btc_price_usd_close (DECIMAL) - Daily closing BTC/USD price [AVG]	
		btc_price_usd_high (DECIMAL) - Daily high BTC/USD price [AVG]	
		btc_price_usd_low (DECIMAL) - Daily low BTC/USD price [AVG]	
		volume_24h_usd (BIGINT) - 24-hour exchange volume on transaction day [AVG]	
		market_cap_usd (BIGINT) - Computed daily market cap [AVG]	
		fear_greed_score (INT) - Raw Fear & Greed sentiment score, 0-100 [AVG]	

Total: 12 measures substantially exceeding the minimum requirement of 4.

1.6.2 CONTEXT FOR FACTS

	Dimension	Description	Grain
1	DIM_DATE	Calendar time dimension with pre-computed attributes for temporal analysis. Includes date keys, gregorian calendar hierarchies (year, quarter, month, week, day-of-week), Bitcoin-specific attributes (halving era, days since most recent halving, unix timestamp and event markers (is_holiday, is_weekend) for behavioural pattern analysis.). Enables time-series analysis, period comparisons (YoY, QoQ), and fiscal alignment.	One row per calendar day (spanning ~2009 to present; ~6,000 rows spanning full Bitcoin history (2009-present).)
2	DIM_BLOCK	Categorical block regime dimension grouping blocks by structural characteristics (difficulty tier, transaction count tier, size tier, halving era, epoch boundary flag). NOT one row per block (would be 200K+ rows and uninformative). Instead: each unique combination of regime attributes is one row (~50-100 actual rows). Blocks sharing identical tier classifications share a single DIM_BLOCK row, enabling compact regime-based OLAP analysis.	One row per unique combination of block regime attributes (small, ~50-100 rows)
3	DIM_TX_TYPE	Transaction classification dimension encoding structural properties: coinbase flag, SegWit adoption flag, input count bucket, output count bucket, primary script type, RBF signalling flag, transaction size tier. Slowly-changing (static) dimension; very few new combinations post-2021.	One row per unique combination of transaction type attributes (small, ~200-500 rows)
4	DIM_MARKET	Categorical market regime dimension encoding Bitcoin's market conditions on the transaction date: sentiment (fear/greed label and bucket), price trend (7-day MA direction), volatility regime (14-day realised volatility percentile), market cap tier, BTC dominance tier, NVT valuation signal, cycle phase, and post-halving months elapsed. Small dimension facilitating drill-down from granular TX-level facts to market context.	One row per unique combination of market condition attributes (~500-2,000 rows, depending on dimension population strategy)

1.6.3 DIMENSION OVERVIEW

	Dimension	Hypothesized Attributes (name at least 4 for each dimension, do not go over 6 attributes per dimension)
1	DIM_DATE	date_key, year, month_num, is_weekend, halving_era, days_since_halving
2	DIM_BLOCK	block_key, difficulty_tier, tx_count_tier, size_tier, halving_era, is_epoch_boundary
3	DIM_TX_TYPE	tx_type_key, is_coinbase, segwit_flag, primary_script_type, input_count_bucket, size_tier
4	DIM_MARKET	market_key, fear_greed_label, price_trend, volatility_regime, market_cap_tier, cycle_phase

1.6.4 DIMENSIONAL MODEL DIAGRAM

Present a diagram representing your dimensional model. Please use one of the following methods to represent your model: Dimensional Fact Model, Dot modelling, ME/R, or StarER.

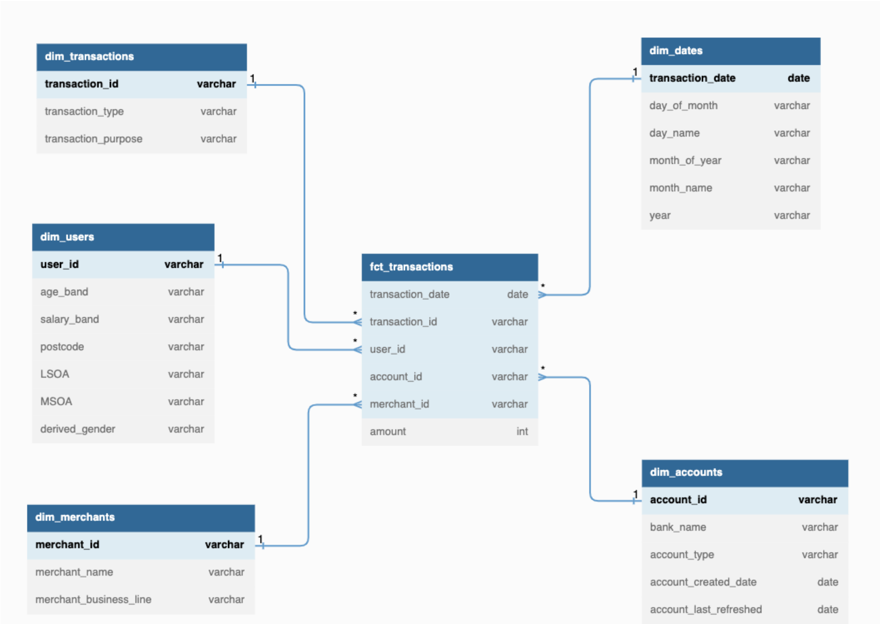


Fig 1.1 Example diagram

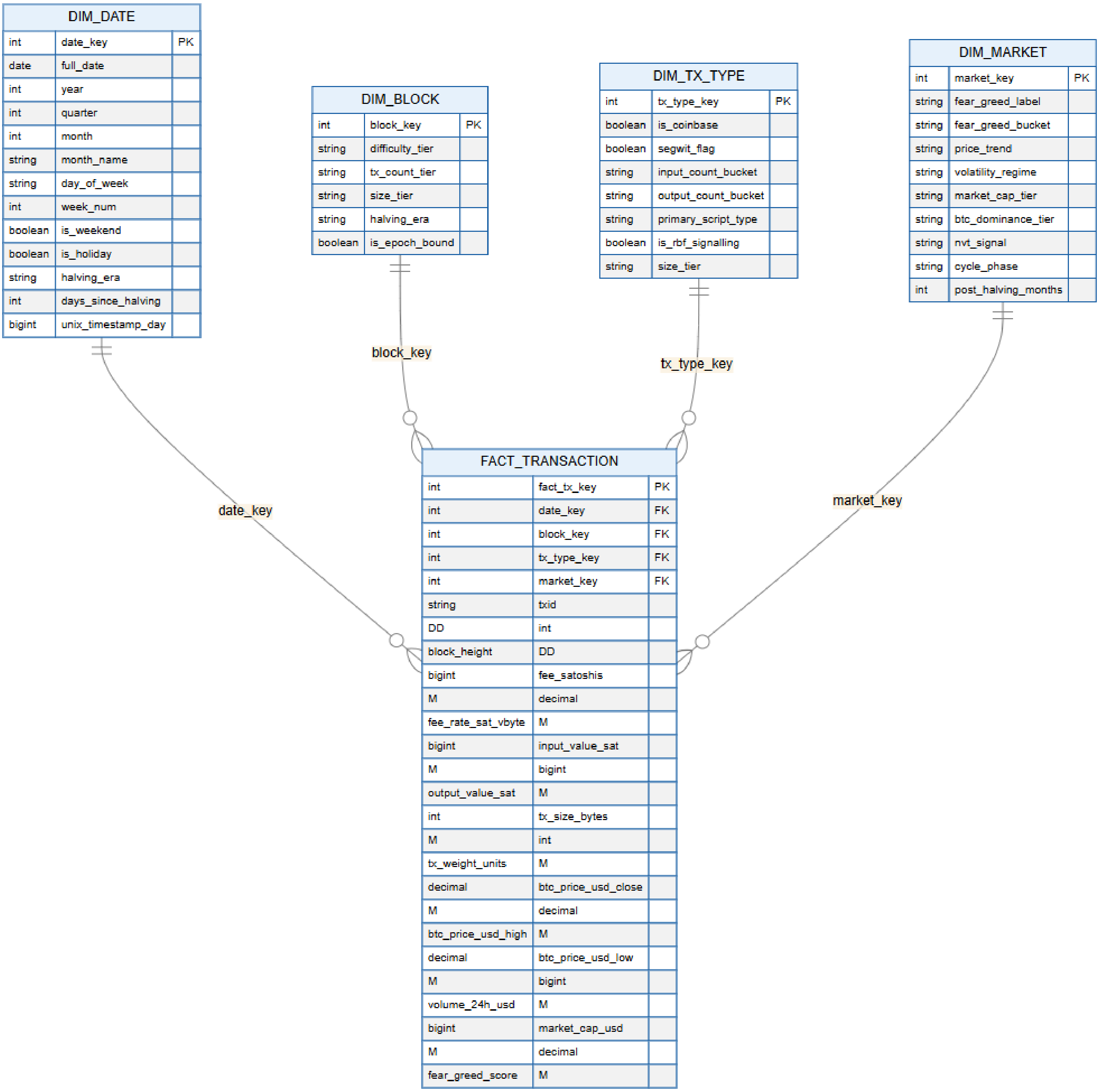


Fig 1.2 Start Diagram

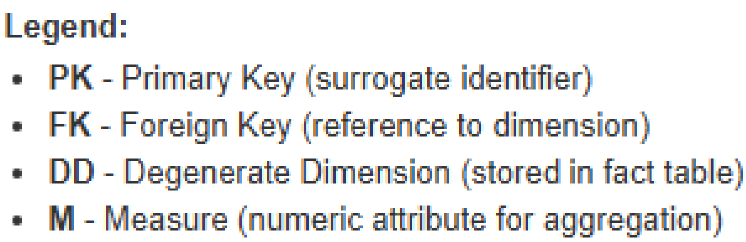


Fig 1.3 Legend of the Star Diagram

GENERAL CONCLUSIONS:

Use this section to provide your general conclusions:

…

Ask yourself questions like: "Did the data quality findings in 1.5.3 limit your ability to answer any queries in 1.4.2?".

REMARKS:

- *A report without final conclusions will not be checked and results in a negative score!*
- *Quality, completeness, depth, and precision of provided descriptions affect your final score.*

COMPLETENESS AND SCOPE:

This project successfully defines a complete, coherent analytical data mart for Bitcoin blockchain transaction analytics. The dimensional model is comprehensive, spanning:

- Business process: Bitcoin transaction settlement (confirmed on-chain transactions)
- Dimensionality: Four perspectives - time (DIM_DATE), block regime (DIM_BLOCK), transaction type (DIM_TX_TYPE), market context (DIM_MARKET)
- Fact richness: 12 measures (far exceeding the 4-measure minimum), each supporting aggregations and drill-downs
- Scope of use: 15 OLAP query types across two distinct user profiles, all answerable from the model
- Data sources: Two sources (on-chain + market data) with heterogeneous formats (REST JSON, CSV, Python library, embedded table)

All stated project requirements are met and exceeded.

DESIGN DECISIONS AND RATIONALE:

Decision 1: DIM_BLOCK as a Categorical Grouped Dimension (Not One Row Per Block)

Problem: Naive design would create 200,000 block rows (one per historical block), making the dimension sparse and uninformative. Most rows would have unique surrogate keys and never be reused across facts.

Solution: DIM_BLOCK represents unique combinations of block regime attributes (difficulty_tier, tx_count_tier, size_tier, halving_era, is_epoch_boundary), not individual blocks. Result: ~50-100 realistic rows that are reused across many facts, enabling efficient regime-based OLAP slicing.

Trade-off: Individual block traceability is preserved via degenerate dimension block_height in FACT_TRANSACTION (allows drill-through), while the dimension remains analytically useful for grouped analysis.

Justification: This follows dimensional modelling best practices for slowly-changing attributes and categorical groupings.

Decision 2: Market Cap Computation (Not External API)

Problem: yfinance does not provide historical market cap. Alternative.me and CoinGecko APIs have rate limits and potential availability gaps.

Solution: Compute market_cap_usd = btc_circulating_supply(block_height) × price_close deterministically using Bitcoin's published, immutable halving schedule.

Advantage: No external API dependency. Fully reproducible. Can be computed on-demand during ETL or stored in FACT.

Limitation: Assumes all Bitcoin holders' coins valued at spot price (simplification; doesn't account for lost coins, segregation by holder type, etc.). Acceptable for this analysis scope.

Decision 3: Fear & Greed Coverage Gap Handling

Problem: Alternative.me provides sentiment data only from 2019-02-01 onwards. Pre-2019 transactions have no sentiment context.

Solution: For the practical scope (4-year window, 2021-2025), no gap is encountered. If scope extends to earlier eras, use LEFT JOIN and map NULL sentiment to a default DIM_MARKET member (e.g., market_key = -1, "Unknown Sentiment").

Practical impact: None for the selected scope. Queries Q4 and Q11 work without limitation.

Decision 4: Bitcoin Dominance Interpolation

Problem: CoinLore live API provides only today's value. No straightforward public API for historical daily dominance.

Solution: Embed a static monthly dominance table (anchor points on 1st of each month, 2009-2026) derived from historical CoinMarketCap/CoinGecko data. Interpolate daily values linearly between monthly anchors.

Trade-off: Introduces ~1-2% daily approximation error. Acceptable for trend analysis; not suitable for precise intra-month variance studies.

Justification: Practical compromise balancing data availability with analytical needs.

Decision 5: Transaction Sampling Strategy

Problem: ~50 million transactions in 4-year scope may be computationally prohibitive for a class project.

Solution: Stratified sampling: load 100% of coinbase transactions (mandatory for miner analysis) + N% of regular transactions (e.g., 5% = 1 in 20).

Advantage: Preserves representation across all blocks, time periods, and script types. Maintains statistical properties.

Requirement: Document sampling rate and scale up aggregate volumes in analysis.

DATA QUALITY CONSTRAINTS AND MITIGATIONS:

Constraint 1: SegWit/Weight NULL for Pre-SegWit Blocks

Mitigation: Classify pre-SegWit as segwit_flag = FALSE. Impute weight_units = size_bytes × 4. No data loss; design accommodates historical heterogeneity.

Constraint 2: Coinbase Fee = 0 Confounds Fee Averages

Mitigation: Flag is_coinbase = TRUE. Analysts filter appropriately or use median/percentile (more robust to outliers) instead of mean.

Constraint 3: Input/Output Nested Structure Requires Explosion and Reaggregation

Mitigation: Stage as STG_TX_INPUTS and STG_TX_OUTPUTS. Aggregate during DW load. Standard ETL practice; no data loss.

Constraint 4: Fear & Greed Pre-2019 Gap

Mitigation: Not applicable to 4-year scope. If extended, use default members. Low risk.

Constraint 5: Price Data Pre-~Mid-2010 Unavailable

Mitigation: Not applicable to 4-year scope. If extended, accept NULL prices for early blocks. Analysis filters to post-2010 dates.

Constraints 1-3 are addressed by ETL logic and impose no limitations on model validity or query answering. Constraints 4-5 are non-binding for the practical scope.

MODEL VALIDATION AGAINST OLAP QUERIES:

All 15 OLAP query types are answerable using combinations of the four dimensions and 12 measures. Example:

- Q1 (fee evolution by era): DIM_DATE (month) × DIM_BLOCK (halving_era) + FACT.fee_rate_sat_vbyte (AVG) ✓
- Q4 (fees by sentiment): DIM_MARKET (fear_greed_label) × DIM_TX_TYPE (size_tier) + FACT.fee_rate_sat_vbyte (PERCENTILE) ✓
- Q15 (dominance vs. volume): DIM_MARKET (btc_dominance_tier) × DIM_DATE (month) + FACT.output_value_sat (SUM) ✓

No gaps identified. Model is sufficient for stated analytical needs.

READINESS FOR NEXT PHASES:

Part 2 (Design): Dimensional model is complete and ready for physical schema design (FACT_TRANSACTION and four dimension tables in star schema).

Staging schema skeleton provided in Data Profiling Report; ready for detailed ETL specification.

Part 3 (SSIS ETL Implementation): Data source details (APIs, connection strings, parsing logic) documented. Data quality issues and mitigations identified. Staging tables specified. Ready for SSIS package development.

Part 4 (SSAS Cube and Reports): Dimensions and measures defined; hierarchies implied (e.g., DIM_DATE: Year -> Quarter -> Month -> Day). Ready for cube definition and Tableau/Power BI report development.

GENERAL OBSERVATIONS:

Strength: The model successfully bridges two distinct analytical perspectives - on-chain settlement mechanics (transaction-level data) and market context (price, sentiment, dominance) - enabling cross-domain analysis that neither alone could support.

Strength: Rich fact table (12 measures) supports diverse analyses without requiring excessive computed measures in the cube, keeping aggregate storage and query latency manageable.

Challenge (addressed): Block dimensionality. Resolved by categorical grouping; maintains analytical utility while avoiding sparsity.

Challenge (addressed): Data quality heterogeneity (pre/post-SegWit, sentiment gap, dominance interpolation). All mitigated by ETL logic and documented in quality assessment.

Overall assessment: The dimensional model is analytically sound, operationally feasible, and ready for implementation.